

BP&Co

Technical Specification & Approval Document

Document Information	Details
Document Type	API Integration Specification
API Version	v1
System	BP&Co Performance Management System
Integration	POS Software ↔ BP&Co
Document Date	10 August 2026
Purpose	Technical review and approval before integration

1. Purpose

This document defines the REST API contract for integrating a third-party billing/POS software with BP&Co. The integration allows the billing system to submit employee sales and productivity metric data to BP&Co and retrieve approved performance slip information for verification and synchronization.

The document is intended for technical review and approval by the third-party billing software team before development and integration.

2. API Scope

The Version 1 API consists of the following endpoints:

#	Method	Endpoint	Purpose
1	POST	/api/v1/external/bulk-sync	Submit bulk performance metric data
2	GET	/api/v1/external/sync-status/{reference_id}	Check processing status of a submitted batch/record
3	GET	/api/v1/external/slips/approved	Retrieve approved performance slips
4	GET	/api/v1/external/metrics	Retrieve active BP&Co metric definitions
5	GET	/api/v1/external/users	Retrieve active BP&Co employee mappings

3. Authentication

All API requests require a valid BP&Co API token.

```
Authorization: Bearer {API_TOKEN}
Accept: application/json
Content-Type: application/json
```

The API token will be generated by BP&Co and provided to the authorized third-party integration. The token must be kept confidential and must not be exposed in client-side browser or public application code.

- HTTPS must be used for all API requests.
- Invalid or missing tokens will return HTTP 401.
- Access is restricted according to the integration/store configuration associated with the token.

4. Base URL

The production base URL will be provided by BP&Co during integration setup.

```
https://{BPANDCO-DOMAIN}/api/v1/external
```

5. API 1 – Bulk Data Sync

5.1 Endpoint

```
POST /api/v1/external/bulk-sync
```

This endpoint accepts multiple employee records and their metric values in a single request. The API validates each employee and metric, creates or updates the corresponding BP&Co slip entries, and calculates performance points according to the BP&Co scoring configuration.

5.2 Request Body

```
{
  "sync_date": "2026-08-10",
  "store_code": "STORE_001",
  "data": [
    {
      "user_identifier": "EMP101",
      "date": "2026-08-10",
      "metrics": [
        {
          "metric_code": "TOTAL_SALES_INR",
          "value": 45000.50
        },
        {
          "metric_code": "BILLS_COUNT",
          "value": 38
        },
        {
          "metric_code": "ITEMS_SOLD",
          "value": 112
        }
      ],
      "reference_id": "POS-BATCH-8821"
    }
  ]
}
```

5.3 Request Fields

Field	Type	Required	Description
sync_date	date	Yes	Date on which the synchronization request is submitted.
store_code	string	Yes	Store/business identifier associated with the integration.
data	array	Yes	Collection of employee metric records.
user_identifier	string	Yes	Employee identifier. Employee code is preferred; mobile/email may also be supported.
date	date	Yes	Performance/slip date for the record.
metrics	array	Yes	List of metric values for the employee and date.
metric_code	string	Yes	Unique BP&Co metric code.
value	number	Yes	Submitted metric value.
reference_id	string	Yes	Unique external reference used for duplicate protection and tracking.

5.4 Example Success Response

```
{
  "success": true,
  "message": "Bulk data sync processed successfully.",
  "summary": {
    "total_records_received": 1,
    "successful_records": 1,
    "failed_records": 0
  },
  "results": [
    {
      "user_identifier": "EMP101",
      "date": "2026-08-10",
      "reference_id": "POS-BATCH-8821",
      "status": "success",
      "metrics_processed": [
        {
          "metric_code": "TOTAL_SALES_INR",
          "value": 45000.50,
          "daily_points_earned": 150.00,
          "slip_id": 1042
        },
        {
          "metric_code": "BILLS_COUNT",
          "value": 38,
          "daily_points_earned": 50.00,
          "slip_id": 1043
        }
      ]
    }
  ],
  "errors": []
}
```

6. Duplicate Protection

The third-party system may retry a request when a network timeout or communication failure occurs. BP&Co will use `reference_id` to identify previously submitted records and prevent duplicate slip creation.

- `reference_id` must be unique for the relevant external record/batch.
- A repeated `reference_id` will not create duplicate records.
- The API will return the existing processing status where applicable.

7. API 2 – Sync Status

7.1 Endpoint

```
GET /api/v1/external/sync-status/{reference_id}
```

This endpoint allows the third-party billing software to verify whether a previously submitted reference has been processed.

7.2 Example Request

```
GET /api/v1/external/sync-status/POS-BATCH-8821
```

7.3 Example Response

```
{
  "success": true,
  "reference_id": "POS-BATCH-8821",
  "status": "processed",
  "total_records": 1,
  "successful_records": 1,
  "failed_records": 0,
  "processed_at": "2026-08-10 11:35:00"
}
```

Supported processing statuses:

- pending
- processed
- partially_processed
- failed

8. API 3 – Fetch Approved Slips

8.1 Endpoint

GET /api/v1/external/slips/approved

This endpoint returns only BP&Co performance slips that have been approved. Pending and rejected slips will not be included.

8.2 Query Parameters

Parameter	Type	Required	Description	Example
date	date	Optional	Specific slip date. Use instead of a date range.	2026-08-10
start_date	date	Optional	Start date for a date range.	2026-08-01
end_date	date	Optional	End date for a date range.	2026-08-10
user_identifier	string	Optional	Employee code, mobile or email.	EMP101
metric_code	string	Optional	Specific BP&Co metric code.	TOTAL_SALES_INR
page	integer	Optional	Page number for paginated results.	1
per_page	integer	Optional	Number of records per page.	100

Date filter rule: use either date OR start_date + end_date. Do not submit both forms together.

Example – single date:

```
GET
/api/v1/external/slips/approved?user_identifier=EMP101&metric_code=TOTAL_SALES_INR&date=2026-08-10
```

Example – date range:

```
GET /api/v1/external/slips/approved?user_identifier=EMP101&start_date=2026-08-01&end_date=2026-08-10
```

8.3 Example Response

```
{
  "success": true,
  "filters": {
    "date": "2026-08-10",
    "user_identifier": "EMP101",
    "metric_code": "TOTAL_SALES_INR"
  },
  "total_records": 1,
  "data": [
    {
      "slip_id": 1042,
      "user": {
        "employee_code": "EMP101",
        "name": "Rajesh Kumar",
        "mobile": "9876543210"
      },
      "metric": {
        "metric_code": "TOTAL_SALES_INR",
        "name": "Total Daily Sales (₹)",
        "unit": "INR"
      },
      "date": "2026-08-10",
      "value": 45000.50,
      "daily_points_earned": 150.00,
      "status": "approved",
      "approved_at": "2026-08-10 11:30:00"
    }
  ],
  "pagination": {
    "current_page": 1,
    "per_page": 100,
    "total": 1,
    "last_page": 1
  }
}
```

9. API 4 – Fetch Active Metrics

9.1 Endpoint

GET /api/v1/external/metrics

Returns active BP&Co metrics available for integration. The third-party system should use metric_code as the stable mapping identifier.

```
{
  "success": true,
  "metrics": [
    {
      "id": 1,
      "code": "TOTAL_SALES_INR",
      "name": "Total Daily Sales (₹)",
      "value_type": "quantity",

```

```
{
  "unit": "INR"
},
{
  "id": 2,
  "code": "BILLS_COUNT",
  "name": "Total Bills Generated",
  "value_type": "quantity",
  "unit": "Bills"
},
{
  "id": 3,
  "code": "ITEMS_SOLD",
  "name": "Total Items Sold",
  "value_type": "quantity",
  "unit": "Items"
}
]
```

10. API 5 – Fetch Active Employees

10.1 Endpoint

GET /api/v1/external/users

Returns active BP&Co employees available for mapping with the third-party billing software.

```
{
  "success": true,
  "users": [
    {
      "id": 10,
      "employee_code": "EMP101",
      "name": "Rajesh Kumar",
      "mobile": "9876543210",
      "role": "Sales Associate"
    }
  ]
}
```

11. Data Mapping Standards

11.1 Employee Identification

The preferred employee identifier is employee_code.

Priority	Identifier	Recommendation
1	employee_code	Preferred and recommended integration identifier.
2	mobile	Supported fallback identifier where required.
3	email	Supported fallback identifier where required.

11.2 Metric Identification

Metrics must be identified using the unique metric_code returned by the metrics endpoint. Internal database IDs should not be used as the integration contract.

```
TOTAL_SALES_INR  
BILLS_COUNT  
ITEMS_SOLD
```

12. Approval and Status Handling

The API will return approved slip information only through the approved-slips endpoint. The third-party system cannot change BP&Co approval status through these APIs.

Slip status is controlled by BP&Co's configured approval process and scoring rules.

Status	Meaning
pending	Record is awaiting approval, where applicable.
approved	Record has been approved and is available through the approved-slips API.
rejected	Record has been rejected and is excluded from the approved-slips API.

13. Validation and Error Handling

HTTP Status	Meaning
200	Request processed successfully.
201	Resource successfully created, where applicable.
400	Invalid request.
401	Missing or invalid API token.
403	Access is not permitted.
404	Requested resource not found.
422	Validation error.
429	Too many requests.
500	Internal server error.

Example validation response:

```
{  
  "success": false,  
  "message": "Validation failed.",  
  "errors": [  
    {  
      "user_identifier": "EMP999",  
      "error_code": "USER_NOT_FOUND",  
      "message": "Employee could not be found."  
    }  
  ]  
}
```

Where possible, validation failures will be returned at item level so that valid records in the same bulk request can continue to be processed.

14. Request and Pagination Limits

- Initial bulk-sync limit: up to 25 employee records per request.
- Large data volumes should be divided into multiple batches.
- Approved slip results are paginated using page and per_page.
- The initial default page size is 100 records.

15. Security Requirements

- All API communication must use HTTPS.
- Bearer API tokens must be stored securely.
- API tokens must not be exposed in public frontend code.
- BP&Co may revoke or disable an API token when required.
- Requests are validated against the integration configuration associated with the token.
- API requests and processing results may be logged for troubleshooting and audit purposes.

16. Integration Sequence

1. The third-party system obtains the API token and BP&Co base URL.
2. The third-party system calls GET /metrics to retrieve available metric codes.
3. The third-party system calls GET /users to retrieve active employee mappings.
4. The third-party system maps its employees and billing metrics to BP&Co identifiers.
5. The third-party system submits daily data through POST /bulk-sync.
6. BP&Co validates and processes the submitted records.
7. The third-party system may use GET /sync-status/{reference_id} to verify processing.
8. The third-party system may call GET /slips/approved to retrieve approved slips and points.

17. Technical Compatibility Confirmation

The third-party technical team is requested to confirm:

- REST API over HTTPS is supported.
- JSON request and response formats are supported.
- Authorization: Bearer token authentication is supported.
- POST requests can be sent for bulk synchronization.
- GET requests can be used for master data, sync status and approved slips.
- The system can generate and retain a unique reference_id for each submitted batch/record.
- Employee and metric mappings can be maintained using the BP&Co identifiers.
- HTTP status codes and validation responses can be handled.
- Paginated responses can be processed for approved slip retrieval.

18. Testing and Acceptance

Before production activation, the following test cases should be completed:

Test	Expected Result
Valid API token	Request is authenticated successfully.
Invalid API token	HTTP 401 is returned.
Valid bulk sync	Records are created/processed successfully.
Invalid employee	Item-level USER_NOT_FOUND error is returned.

Invalid metric	Item-level INVALID_METRIC error is returned.
Duplicate reference_id	Duplicate slip creation is prevented.
Approved slip query	Only approved slips are returned.
Pending/rejected slip query	Pending/rejected slips are excluded.
Date range query	Records within the requested date range are returned.
Metric filtering	Only the requested metric records are returned.
Employee filtering	Only the requested employee records are returned.
Pagination	Results are returned correctly across pages.

19. API Versioning

Version 1 endpoints use the /api/v1/external/ path. Any future breaking API changes can be introduced through a new version without changing the existing V1 contract.

```
/api/v1/external/...
```

20. Technical Approval

Confirm that the API specification has been reviewed and is technically acceptable for integration.

End of Document